MATTEO SAPONATI

Research Scientist - Project Lead

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Zürich, Switzerland

Experience

Research Scientist
2026 - present

Tufa Labs, Zurich (CH)
Full-time, on-site

- Designed pretraining and synthetic data pipelines to build very small, highly capable reasoning models.
- Developed and tested automatic ML research workflows with coding agents.
- Led research projects on self-distillation post-training algorithms and supervised interns.

Consultant - Neuromorphic Computing
2025 - 2025

Neuronova, Milan (IT)
Part-time, remote

- Conducted an R&D project on Edge AI for audio applications.
- Provided expert consulting on neuromorphic training algorithms and hardware–software codesign principles
- Worked independently as an external consultant, coordinating with the Neuronova team throughout the project lifecycle.

Postdoctoral Research Scientist
2023 - 2025

Institute of Neuroinformatics, ETH/UZH, Zurich (CH)
Full-time, on-site

- Secured competitive research funding to design advanced training algorithms for edge AI.
- Led research and deployment of Large Language Models, Spiking Neural Networks, and Recurrent Neural Networks.
- Published 4 peer-reviewed papers; presented at 8 international conferences.
- Supervised M.Sc. and Ph.D. students from ETH Zurich, University of Zurich, and ZHAW Center for Artificial Intelligence.

PhD researcher
2020 - 2023

Max-Planck Institute for Brain Research and Ernst Strüngmann Institute for Neuroscience, Frankfurt Am Main (DE)
Full-time, on-site

- Designed learning algorithms for Spiking Neural Networks applied to Machine Learning and Neuroscience.
- Published 3 peer-reviewed articles; presented at 6 international conferences.
- Led data analysis projects; Coordinated scientific seminars.

Assistant Research Scientist
2019

Institute des Neurosciences des Systèmes, Aix-Marseille University, Marseille (FR)
Full-time, on-site

- Conducted research in Computational Neuroscience.
- Developed models to describe neuronal coupling across spatial scales.
- Investigated theoretical limits using The Virtual Brain platform (TVB).

Research Intern
2018

Barcelona Biomedical Research Park, Barcelona (ESP)
Full-time, on-site

- Published 2 papers and presented research at international conferences.

Education

2020 - 2023 **Ph.D. in Neuroinformatics**
Highest Honors (top 5%) - Donders Centre for Neuroscience, Radboud University (NL)

2016 - 2018 **M.Sc. in Physics**
110/110 - Department of Physics, University of Pisa (IT)

2011 - 2016 **B.Sc. in Physics**
94/110 - Department of Physics, University of Pisa (IT)

Skills

Coding Skills Python, PyTorch, Codex/Claude Code, Linux, LaTex, C++

Research Skills Machine Learning, LLM training and inference, Mechanistic Interpretability, Mathematical Modelling, Critical Thinking, Public Speaking, Teamwork, Problem Solving

Language Skills Italian (Mother tongue), English (Business Fluent), Portuguese (Business Intermediate), French (Basic)

Research

Haag, J., Metzner, C., Zendrikov, D., Indiveri, G., Grewe, B., De Luca*, C., & **Saponati*, M.** (2026). Mixed-signal implementation of feedback-control optimizer for single-layer spiking neural networks. *IEEE International Symposium on Circuits and Systems*. https://doi.org/10.48550/arXiv.2603.24113

Saponati, **M.**, & Vinck, M. (2026). Inhibitory feedback enables predictive learning of multiple sequences in neural networks. *Neural Computation*, *38*(4), 471–498. https://doi.org/https://doi.org/10.1162/NECO.a.1504

Stalder, N., Grewe, B., **Saponati***, **M.**, & Aceituno Vilimelis*, P. (2026). A combination of noise and bilateral filters achieve supralinear and scalable adversarial robustness in cnns. *The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026*. Retrieved June 3, 2026, from https://cvpr.thecvf.com/virtual/2026/poster/37609

Saponati, **M.**, De Luca, C., Indiveri, G., & Grewe, B. (2025). A feedback control optimizer for online and hardware-aware training of spiking neural networks. *2025 Neuro-Inspired Computational Elements Conference (NICE)*. https://doi.org/10.1109/NICE65350.2025.11065428

Saponati*, **M.**, Sager*, P., Aceituno Vilimelis, P., Stadelmann, T., & Grewe, B. (2025). The underlying structures of self-attention: Symmetry, directionality, and emergent dynamics in transformer training. *Proceedings of the 41st International Conference on Machine Learning*. https://doi.org/10.48550/arXiv.2502.10927

Saponati, **M.**, & Vinck, M. (2023). Sequence anticipation and spike-timing-dependent plasticity emerge from a predictive learning rule. *Nature Communications*, *14*(1), 4985. https://doi.org/10.1038/s41467-023-40651-w

Saponati, **M.**, Garcia-Ojalvo, J., Cataldo, E., & Mazzoni, A. (2022). Thalamocortical Spectral Transmission Relies on Balanced Input Strengths. *Brain Topography*, *35*(1), 4–18. https://doi.org/10.1007/s10548-021-00851-3

Spyropoulos, G., **Saponati**, **M.**, Dowdall, J. R., Schölvinck, M. L., Bosman, C. A., Lima, B., Peter, A., Onorato, I., Klon-Lipok, J., Roese, R., Neuenschwander, S., Fries, P., & Vinck, M. (2022). Spontaneous variability in gamma dynamics described by a damped harmonic oscillator driven by noise. *Nature Communications*, *13*(1), 2019. https://doi.org/10.1038/s41467-022-29674-x

Saponati, **M.**, Garcia-Ojalvo, J., Cataldo, E., & Mazzoni, A. (2019). Integrate-and-fire network model of activity propagation from thalamus to cortex. *Biosystems*, *183*, 103978. https://doi.org/10.1016/j.biosystems.2019.103978

Grants and Awards

Jan 2024 - Jan 2026 **ETH Postdoctoral Fellowship**
ETH Zurich Postdoctoral Fellowship programme (Zürich, CH)

Mar 2023 **Cosyne Presenters Travel Grant**
Cosyne Conference 2023 (Montreal, CA)

Sep 2019 - Sep 2023 **IMPRS Research Fellowship**
International Max Planck Research School (IMPRS) for Neural Circuits, MPI for Brain Research, Frankfurt am Main (DE)

Jul 2018 - Aug 2018 **Erasmus+ Grant**
Erasmus program (EU)